

Final Exam Solution - SOC2101: Soc., Tech. and Engg. Ethics

Q1: Analysis using the ACM Code of Ethics

John works in a company manufacturing drugs for painless death, primarily for a country where euthanasia is legally supported. The practice now includes individuals with severe depression.

Analysis of ACM Code of Ethics:

1. Public Welfare (1.1): The ACM Code emphasizes the responsibility to contribute to the well-being of society. The manufacture of euthanasia drugs raises concerns about the impact on vulnerable individuals and societal values.
2. Honesty (2.3): The claim that the practice is painless cannot be verified until after death, which might be misleading. Being honest about the limitations of the drug is crucial.
3. Privacy (1.6): Euthanasia in the context of depression involves sensitive health data. Ensuring that privacy is maintained and that individuals' data is protected is a key ethical consideration.
4. Fairness (1.7): The law supports euthanasia in this country, but the ethical concerns around the reform to include people with severe depression require careful consideration. The decision must be based on thorough, unbiased assessments.
5. Quality (2.1): The company should ensure the quality and safety of the drug, ensuring it's administered in a way that aligns with both ethical and medical standards.

Q2: Ethical Violations in Machine Learning System

The university uses machine learning to predict student performance and identify at-risk students, utilizing data from attendance, grades, extracurriculars, and social media behavior.

Ethical Violations:

1. Privacy (1.6): Using social media data for academic decisions raises significant privacy concerns. Students may not have consented to their data being used in this way.
2. Fairness (1.7): The system might perpetuate biases, especially if social media data is biased or inaccurate. Predictions may be unfairly skewed based on factors unrelated to academic performance.
3. Bias (1.7): Machine learning systems can inherit biases from the data they are trained on. If certain groups are underrepresented or misrepresented, predictions may be biased.
4. Unintended Consequences (1.1): Over-reliance on automated predictions could lead to unjust consequences, such as misidentifying students who are at risk of academic probation.

Alternative Approaches:

- Prioritize transparency and consent for data usage, ensuring students are aware of how their data is being used.
- Use non-sensitive data and ensure that machine learning systems are regularly audited for fairness.
- Focus on human oversight, allowing faculty to intervene when predictions are made.

Q3: Ethical Concerns in Instagram's "Project Uplift"

Instagram conducted a study by tracking users' sadness and using AI to generate uplifting content.

Ethical Concerns:

1. Informed Consent (1.6): Users were unaware that their data was being tracked and used for psychological experiments. Proper consent should have been obtained before using their data.
2. Privacy (1.6): Gathering and analyzing personal content such as emotional posts without explicit permission is a violation of user privacy.
3. Manipulation (1.4): The AI intervention in users' emotional state could be seen as manipulative, especially without transparency regarding how their emotions were influenced.

Measures Instagram Should Have Taken:

- Obtain clear and explicit consent from users before collecting sensitive data.
- Be transparent about how the data would be used and the purpose of the research.
- Ensure that the experiment does not harm users psychologically and provide support in case users feel uncomfortable with the results.

Q4: Framework for Handling User Data in the App

Mariam, a computer engineer, works on a medical app that stores users' sensitive health data. The marketing and analytics departments request access to this data.

Proposed Framework:

1. Data Minimization (1.6): Only collect and share data that is necessary for the specific purpose (e.g., aggregated usage data for analytics).
2. Informed Consent (1.6): Users should be fully informed about how their data will be used, including any sharing with marketing or analytics departments.
3. Transparency (2.3): Ensure that users know what data is being collected and how it will be shared. Implement a clear privacy policy and provide users with an option to opt out.

4. Data Protection (1.6): Implement strong security measures to protect users' sensitive health data and prevent unauthorized access.
5. Ethical Review (1.1): The company should have an ethical review process for any data-sharing requests, ensuring they align with privacy and ethical standards.

Q5: Environmental Impacts of Superhero Technology

As a member of S.H.I.E.L.D., tasked with reducing the environmental impact of superhero technology, you must consider the environmental consequences of gadgets and technology.

Environmental Impacts:

1. E-Waste: Superhero gadgets and devices might quickly become obsolete, contributing to significant electronic waste if not disposed of properly.
2. Energy Consumption: Advanced technologies, like energy-intensive gadgets, could have high carbon footprints.
3. Material Sourcing: Mining for rare materials used in advanced tech could result in environmental degradation.

Mitigation Strategies:

1. Eco-friendly Materials: Use sustainable materials in the construction of gadgets and ensure that they are recyclable at the end of their life cycle.
2. Energy Efficiency: Focus on developing energy-efficient devices that require less power to operate and are more durable.
3. Circular Economy: Implement a take-back program to reclaim old gadgets for reuse or recycling, reducing the need for new materials.
4. Green Technologies: Incorporate renewable energy sources and reduce the carbon footprint of manufacturing processes.
5. Cost-Effectiveness: Choose cost-effective, environmentally friendly alternatives that reduce long-term waste disposal costs and minimize the impact on the environment.